

AI Adoption Barriers in Brokerage Firms: A Systematic Literature Review

Mitchell de Vries

Senior Software and AI Engineer at DMA One

mitchelldevries2001@gmail.com

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Abstract

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1. Introduction

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2. Brokerage Context

Most people have bought a stock at some point, or at least know someone who has. What they probably don't think about is. What happens after they press the button. The trade doesn't just appear in their account, it passes through a chain of licensed firms, clearing systems, and legal processes before ownership actually changes hands. The brokerage is where that chain starts.

Let's put it simply, a brokerage is a licensed intermediary that executes trades on behalf of clients who cannot access financial exchanges directly. The NYSE, NASDAQ, and equivalents don't let just anyone connect to their systems. Exchange membership requires regulatory approval, significant capital reserves, and ongoing compliance obligations that most investors don't want. A brokerage provides those services for a commission.

That much is common to all brokerages. Where they diverge is in who their clients are, and that divergence turns out to matter enormously for everything from regulatory obligations to operational infrastructure. The industry breaks into two distinct models: retail and wholesale.

2.1 Retail Brokerage

Retail brokerages serve private individuals, people investing personal savings through a mobile app or online platform. The typical retail client buys shares in a company, maybe some ETFs, occasionally commodities[X]. Transaction sizes are modest. On the other hand the amount of clients a broker has is enormous.

Because these clients are private individuals with no assumed financial expertise, they get the strongest regulatory protections available. Under MiFID II, brokerages must assess the suitability and appropriateness of investments before providing them, and warn or restrict clients where a product is not appropriate [1]. The basic idea is protective on purpose: small everyday investors are seen as more at risk, in a way that big professional investors are not. A place where retail investors can start trading is Fidelity¹, LYNX², and Easybroker³.

¹<https://www.fidelity.com/>

²<https://www.lynx.nl/>

³<https://www.easybroker.nl/>

2.2 Wholesale Brokerage

Wholesale brokerage is a different business serving a different world. The clients here are not individuals. They are asset managers running billion-euro portfolios, hedge funds executing complex arbitrage strategies, pension funds managing retirement savings at scale, and banks trading on their own account. These are professional counterparties, and they are treated as such.

Wholesale clients are permitted to trade across the full range of available instruments: equities, fixed income, derivatives, structured products, and more without the product restrictions that apply in retail. The assumption is that they understand what they're doing and can absorb the associated risks. This doesn't mean there's no regulation (there is, and it's substantial), but the focus shifts away from protecting clients from themselves and toward market integrity, systemic risk, and operational standards instead.^[x]

Beyond execution, wholesale brokerages typically offer a broader service stack: proprietary research, direct market access, and hands-on client support tailored to the specific needs of institutional accounts. DMA ONE⁴ and Liquidnet⁵ are examples of firms operating in this segment. Table 1 lays out the key contrasts between the two retail and wholesale brokerage.

Table 1: Key differences between retail and wholesale brokerage

	Retail	Wholesale
1	High quantity of transactions	Low quantity of transactions
2	Highest level of regulatory client protection	Reduced regulatory client protection
3	Low individual transaction value	High individual transaction value
4	Restricted to certain investment products	Access to the full range of investment products

The differences in the table are about the client relationship. What can the client do and how does a broker service the client. When a client makes a trade retail or wholesale a post-trade machinery kicks in. Both types of brokerage settle trades the same way. Which raises the obvious question: what does settlement actually involve?

2.3 The Settlement Process

A trade is not finished when it's executed. Execution just means two parties have agreed on a price. The securities still need to change hands, and so does the money. Until both of those things happen, neither side has fully delivered on their side of the deal. The process of making that happen is called settlement.

More precisely, settlement is the exchange of securities for payment, formally known as *Delivery versus Payment* (DvP). The two legs are linked so that neither party is left

⁴<https://www.dma.world/one>

⁵<https://www.liquidnet.com/>

exposed, a principle set out by the central-bank community in the early 1990s [2]. Weiss [3] describes the whole settlement circuit as the “plumbing” of financial markets. That comparison is accurate. Settlement is what actually moves the cash and the securities between parties after a trade is agreed, and like real plumbing it sits in the background where nobody looks at it.

Getting from trade agreement to completed settlement involves several layers of infrastructure. The most important piece to understand is the *Central Counterparty Clearing House*, or CCP. When a trade is executed, the CCP inserts itself in the middle, it becomes the buyer to every seller, and the seller to every buyer. If one counterparty defaults, the CCP covers the other side. The other party gets made whole regardless. CCPs also *net* positions across participants: instead of settling each individual trade, they calculate what each firm owes or is owed across all of its activity in a given instrument and settle a single net figure. For retail brokers routing thousands of small client orders this is efficient, although the size of the netting benefit depends on how concentrated clearing is across CCPs [4].

While wholesale trades utilize this same CCP infrastructure, the post-trade pipeline for institutional market participants is far more fragmented. Institutional trades pass through a more complex chain of custodians holding client securities, prime brokers providing financing and clearing, correspondent banks managing cross-border cash, central securities depositories (CSDs) maintaining the official ownership records. Each of these intermediaries runs its own systems. Getting them all to agree on what needs to move, when, and to whom requires coordination that is partly automated and partly manual. [5] covers this infrastructure in detail. The consequence of all this complexity is straightforward: more things can go wrong, and when they do, a human usually has to fix it.

The settlement lifecycle itself breaks into four phases, regardless of brokerage type:

1. **Trade Capture:** Immediately after execution, the details: instrument, price, quantity and counterparty are recorded.
2. **Confirmation and Matching:** Both sides of the trade independently submit their record of what was agreed. If the two records match, the trade can proceed. If they don't, there's a discrepancy that has to be resolved before anything else can happen.
3. **Clearing:** Net obligations are calculated across all matched trades, typically with a CCP stepping in to guarantee delivery and reduce the bilateral exposure between counterparties.
4. **Settlement:** The actual transfer of securities to the buyer, cash to the seller, legal ownership updated. The trade is done.

Figure 1 illustrates how these phases connect in practice for a typical wholesale trade, from the initial order through to final settlement. What the diagram makes visible is the bilateral nature of the process: both sides independently reconcile their books, custodians operate on each end, and the central depository sits between them coordinating the actual transfer.

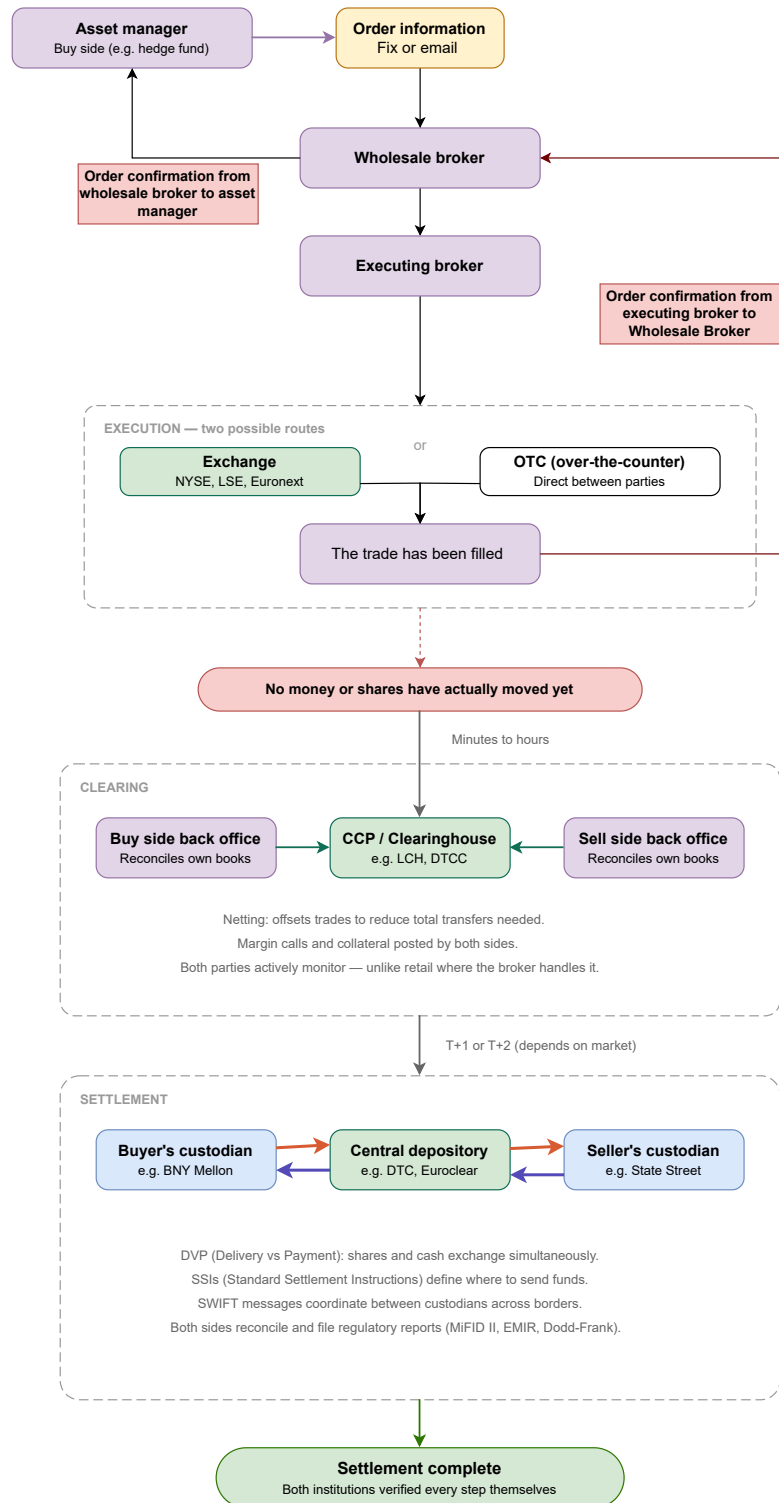


Figure 1: The wholesale settlement process, from order initiation through execution, clearing, and final settlement.

In wholesale brokerage, the gap between confirmation and final settlement is where most of the day-to-day operational difficulty sits. Trades that fail to match, or that don't settle on their intended date, are called *fails* or *late settlements*. Resolving them involves manual investigation, communication between multiple intermediaries, and often time pressure, and chains of unresolved fails can spread to other participants and affect overall market liquidity [6]. Settlement windows in Europe are typically T+2 [7] [8], meaning two business days after execution, and missing that window triggers financial penalties under regulations like CSDR. For trades worth hundreds of millions, a fail is not a small problem. It can mean penalty interest, mandatory buy-ins, and counterparty risk that lands on the firm directly[X].

This is why settlement operations feature so prominently in any survey of AI in wholesale brokerage[X]. The manual workload is high, the stakes are significant, and the processes are structured enough that automation is plausible. The following sections examine where AI is being applied, starting with client-facing functions and moving into the operational back office.

3. AI in Client Lifecycle

Before a wholesale brokerage settles or send trade details to the executing brokerage a significant amount of work has already been done. The client is identified, verified, risk-assessed and on-boarded. This process can become quite complex for institutional clients. They have for example complex legal structures or cross-border operations. Furthermore once the client has been on boarded the brokerage must continuously monitor the client. Updating risk profiles and ensure ongoing compliance. Apart from all this the brokerage is also providing a service too the client. The client expects to receive client support, research about different securities and monitoring of the trades. In this section we will examine the different AI tools that are already being used by wholesale brokerage to improve the client life cycle. We will divide the client life cycle in 4 parts. On boarding, Client monitoring and Client support

3.1 Onboarding

Client on boarding is an important entry point for financial institutions. The client on boarding is there to establish a relationship with new clients. And after that the financial institutions need to maintain this relationship. These tasks can range from operational to regulatory. In the past these tasks have relied heavily on manual work by different people in a company and different teams. On boarding a client and receiving the correct papers mainly comes from the regulators. What a client needs to provide and what laws a brokerage firms needs to adhere too. Depending on the jurisdiction they are in. Brokerage

firms in The Netherlands have different laws than brokerage firms in South Africa.

But let's look at examples of how an onboarding flow looks like for a brokerage firm.

1. **Pre-onboarding scoping:** Before any paperwork starts, the brokerage and the client agree on what the relationship will look like. Which markets the client wants to access, which instruments they will trade, and which services they need (DMA, research, settlement, financing). The effort of the rest of the onboarding scales with this. A pension fund trading equities looks very different from a hedge fund running cross-border derivatives, and the brokerage already adjusts its expectations here.
2. **Document collection:** The brokerage requests a full document pack from the client. For institutional clients this list is long: articles of incorporation, authorized signatory lists, beneficial ownership disclosures (the UBOs), tax forms such as FATCA W-8BEN-E or CRS self-certifications, proof of regulated status where applicable, investment mandates, and source of funds or source of wealth statements. For a fund structure this can multiply quickly. A single asset manager may run dozens of sub-funds, each one a separate legal entity with its own document pack.
3. **Identity and entity verification:** Two things have to be verified separately. The entity itself, and the natural persons authorized to act on its behalf. The entity is checked against company registers, Legal Entity Identifier (LEI) databases, and regulatory registers. The authorized signatories are verified individually with passport, proof of address, and sometimes biometric checks. Beneficial owners (anyone holding more than 25% of the entity, usually) also need to be verified.
4. **Screening:** Once the parties are identified, they are screened against sanctions lists (OFAC, EU, UN), politically exposed person (PEP) lists, and adverse media. The screening applies to the entity, its directors, its authorized signatories, and its beneficial owners. For a corporate client with a layered ownership tree, this can mean dozens of individual screening checks for one onboarding.
5. **Risk classification:** After the data is collected, the brokerage assigns a risk rating to the client. This rating decides how much further due diligence is needed. Inputs are usually jurisdiction (some countries are higher risk by default), entity structure (opaque ownership or special purpose vehicles raise the risk), business activity, expected transaction volumes, and the outcome of the screening. High-risk clients are routed into Enhanced Due Diligence (EDD), which means deeper investigation and senior compliance sign-off. Low-risk clients pass through a simpler track.
6. **Compliance review and sign-off:** A compliance officer reviews the file, signs off on the risk assessment, and either approves the client, requests more information,

or rejects the application. For institutional clients with complex structures this review can take weeks. It often loops back to the client several times to supplement or correct the documentation. According to **friday2022onboarding**, on boarding for cross-border institutional clients historically takes four to six weeks, sometimes longer.

7. **Account setup and activation:** Once approved, the brokerage opens the accounts in its internal systems, sets up settlement instructions (where securities and cash should be delivered), provisions market access, configures users, and activates the client. From here the relationship moves into the monitoring phase, which we cover in the next subsection.

This is the manual version of the flow, and it is also the version where AI has the clearest entry points. Document collection, identity verification, screening, and risk classification are repetitive and rules-driven enough that they can be partly or fully automated. Let's look at what wholesale brokerages are right now using AI to make the flow more efficient.

The previous flow describes the manual version of on boarding. The version below describes the same steps but with the AI tools that are actually deployed today. For each step we look at why AI helps, how it does it, and who is using it. Some steps are mature and widely adopted. Others are still experimental or barely touched. The maturity is uneven across the flow, and that unevenness is itself part of the picture.

1. **Pre-onboarding scoping:** This step is the least AI-touched of all the stages. The conversation about scope, services and instruments is still mostly human. Some brokerages use predictive analytics on prospect data to flag which prospects are likely to be a good commercial fit, and AI-guided conversational assistants are sometimes used to help clients through the initial digital intake. But the bulk of AI value sits downstream.
2. **Document collection:** This is one of the more mature use cases. OCR combined with NLP, often packaged together as Intelligent Document Processing (IDP), reads articles of incorporation, tax forms, authorized signatory lists, and beneficial ownership disclosures, and pulls the structured fields out of them **idp2025banking**. Form logic engines also adjust the document pack required based on entity type and jurisdiction, so a low-risk client in a clean jurisdiction does not get asked for the same depth of information as a high-risk one **friday2022onboarding**. Vendors active in the financial sector here include Quantiphi (Dociphi) and Synapse Analytics (Doxter).
3. **Identity and entity verification:** AI-driven document analysis matches the photo ID submitted by an authorized signatory to the underlying document, and biometric

facial recognition matches the live person in front of the camera to that ID. For entities, AI agents pull from LEI registers, national company houses, and regulator databases automatically rather than relying on a compliance officer to do the lookup. Sumsub and Trulioo are the most commonly named vendors for KYC and KYB in the financial sector. The output of this step is essentially a yes/no on whether the entity and the people behind it are who they claim to be.

4. **Screening:** Sanctions and PEP screening done with traditional rule-based string matching produces a very high false positive rate, and machine-learning methods are increasingly used to reduce it [9]. This is because legitimate clients often share name fragments with sanctioned individuals, especially across non-Latin scripts. ML-based name matching reduces the false positives, and NLP-based adverse media monitoring scans news in multiple languages and flags any negative coverage connected to the parties. ComplyAdvantage is the most prominent vendor in this space and reports cost reductions of up to 50%, although that figure is from their own marketing. Either way, this is one of the steps where AI clearly outperforms the manual baseline.
5. **Risk classification:** Supervised ML models score clients on jurisdiction, entity structure (opacity, use of SPVs, ownership depth), expected behavior, and the output of the screening step. The score then drives which on boarding track the client goes onto: low-risk fast-track, standard, or Enhanced Due Diligence. Synapse Analytics' Konan is one of the named risk-scoring tools used in financial on boarding. However it is also the step where the most concerns about explainability and bias come up. A risk score that cannot be explained to a client or a regulator is a problem, and we come back to this in the Limits and Risks section.
6. **Compliance review and sign-off:** The newer use case here is agentic AI. AI agents identify missing or inconsistent documents, communicate directly with the client to request what is needed, and only escalate to a human compliance officer when the case actually needs human judgment. According to a 2025 case study from Nagent AI, an AI agent deployed for a wholesale bank's on boarding workflow saved relationship managers 3 to 4 hours per day on document chasing and follow-up communication **nagent2025wholesale**. The review itself, the actual sign-off decision, is still done by a human. What the AI changes is how much of the file is already clean by the time it reaches the human.
7. **Account setup and activation:** The final step is closer to Robotic Process Automation (RPA) than to learned AI. Bots create records in the CRM, set up settlement instructions, provision market access, and configure user accounts across the brokerage's internal systems. RPA is rules-based and not really "AI" in the modern

sense, but it slots into this step well and is often deployed alongside the AI tools in the earlier stages **friday2022onboarding**. Vendors active in this space include HighRadius for B2B financial on boarding and WonderBotz for high-volume account creation. Once this step is done the client is live, and the relationship moves into ongoing monitoring.

What this flow shows is that AI adoption in on boarding is not a single tool but a stack of them. Document processing and screening are mature. Risk scoring and agentic review are newer and still raise governance questions. And the very first and last steps (scoping and account setup) are barely AI-driven at all. The next subsection looks at what happens once the client is live, which is where the AI focus shifts from verification to monitoring.

3.2 Client monitoring

Client monitoring is a time consuming task that can be overlooked quite easily. The biggest reason a brokerage firm needs to do client monitoring is because of regulatory standpoint. A few client monitoring tasks that needs to be happening according to [10] are: MiFID II, MiFIR, MAR (Market Abuse Regulation), and the AML directives (currently AMLD5/AMLD6, moving to the new EU AML Regulation). The only real tasks for brokerage for client monitoring is to monitor the counterparty credit and exposure. Furthermore a different

3.3 Client Support

Wholesale clients don't ask the same kinds of questions as retail clients. A retail client might ask why their app is not loading or how to reset a password. An institutional client is asking things like whether a US fund can buy Chinese A-shares through the broker, what documentation a Polish regulator needs for a derivative trade, or what the cut-off time is for the Japanese market today. These questions are operational and jurisdictional, and getting them wrong has consequences. If a desk gives the wrong settlement cycle for Brazil, the trade fails and somebody pays a penalty.

To handle this, wholesale brokerages run dedicated information service teams. These are specialists who know specific markets, settlement procedures, country-specific regulations, and the tax forms required for each jurisdiction. A large broker might have separate desks for the Americas, Asia, and EMEA, with overlapping coverage so that someone is available during every market's trading hours. This is expensive. The work requires experienced people, and the knowledge mostly sits in their heads, in old emails, and in scattered internal documentation.

This is where AI is starting to be useful. A common application is a chatbot or AI assistant that answers routine market access and settlement questions by retrieving from internal

knowledge bases, a technique known as retrieval-augmented generation [11]. If a client asks what the settlement cycle is for South Korean equities, or what tax form a German pension fund needs before trading US treasuries, the system can answer in seconds. What this means is that the information service team is no longer the first line of contact for every question. They become the second line, dealing with the harder cases where a human still needs to think. On top of that there is a real cost benefit. Maintaining global information service teams across multiple time zones is one of the larger operational costs in wholesale brokerage, and an AI assistant that handles routine queries reduces the need for round-the-clock human coverage in every region.

However, this is also one of the less mature use cases. The problem is the cost of being wrong. Large language models can hallucinate, producing fluent and confident answers that are factually incorrect or unsupported [12]. If a chatbot tells a client that a market has T+2 settlement when it is T+1, the client may act on that and the trade fails. In a wholesale context the trade size is large, so the penalty is large. This is why most firms currently deploy these tools internally first, as an aid to the information service team rather than directly to clients. A human still sees the answer before it goes out.

4. AI in Settlement Operations

Client facing AI applications attract a lot of attention. Because they give you a visual representation into future technologies. But the backbone of wholesale brokerage has an equally important domain for AI adoption which is, compared to the client life cycle underexplored in the literature. This backbone is trade settlement, reconciliation and fail management.

4.1 Trade confirmation and matching

The first step after a trade execution is the confirmation and matching. As a client you are able to buy or sell a security. The counter party will then send an email with a confirmation about the trade or they will send a SWIFT message. A SWIFT message is a standardized electronic message used by financial institution to communicate trade orders. The instructions of the counter party have to match with the instruction on the client side. In chapter 1 we covered the client lifecycle. In this chapter we are going to cover the operational side of the back office. How and where the securities actually switch places

Markets are moving from T+2 to T+1. The US did this on 28 may 2024 [X]. The EU has confirmed a move in October 2027 [X]. This means that the window between a trade execution and a settlement is now one working day. When there is T+2 matching could happen overnight in batches. With T+1 it has to happen intraday or close to real-time.

For example a trade executed at 16:00 on the New York Stocks Exchange(NYSE) used to be able to be matched at the end of the day. With T+1 it has to be affirmed by 21:00 on trade date.

The traditional matching process is a mix of channels. But the most common are: SWIFT messages, FIX, Bloomberg messages or email attachments. For the latter 2 the backoffice has to manually match the trades[X]. Different counterparties use different formats. This a broker may communicate differently with different clients. Here we can already see the first use case of NLP-based extraction. AI models are going to read unstructured confirmation (email bodies, attached PDFs, Bloomberg chat) and pull out the structured fields needed for matching. The fields needed are: instrument, quantity, price, settlement date, counterparty references. A company that is already doing this is Xceptor[X]. ClearDox does something similar but more centered at commodity confirmations. Once the data is structured it will be fed into a central matching utility, normally this is a central security deposit (CSD).

A CSD is the central infrastructure that sits behind a securities market providing clearing, settlement, and central matching services for most equity and fixed income trades not all trades are matched by the CSD. Trades that are not matched by the CSD are mostly traded on an alternative venues that self clear or very illiquid name. As seen in table 2 there are different CSD's for different market's.

Table 2: Major Central Securities Depositories by market

Market	CSD
United States	Depository Trust Company (DTC, part of DTCC)
Belgium / France / Netherlands	Euroclear
Germany / Luxembourg	Clearstream
United Kingdom	Euroclear UK & International (CREST)
Japan	Japan Securities Depository Center (JASDEC)
Hong Kong	Hong Kong Securities Clearing Company (HKSCC)
China	China Securities Depository and Clearing Corporation (ChinaClear)

Every CSD has their own software to match trades. This software is a rules-based central matching utility, and its value comes from operational engineering and network effects rather than machine learning. But the layer in front of it, the bit that turns a PDF into a structured CTM submission, is where AI actually plays a role.

The second use case is error handling. When two records do not match, somebody has to figure out why. ML-based classifiers learn from historical error patterns and route exceptions to the right team without manual triage [X]. Some platforms also use AI to highlight clauses in OTC confirmations that deviate from a standard template, so

a reviewer is pointed at the lines that need attention rather than reading the whole document [X]. This is closer to assisted review than full automation.

As we can see some aspects are more developed than others. Goldman Sachs reported a reduction in same-day affirmation exceptions and settlement fails in the quarter after rolling out CTM's Match-to-Instruct workflow [X], and buy-side adoption of the same workflow grew 91% in 2023 [X]. The AI layer sitting on top of those utilities is newer. NLP-based confirmation extraction works well for structured asset classes like cash equities. It still struggles with bespoke OTC documents where the language is non-standard and the cost of mis-parsing is large. Most wholesale firms today run these tools in a hybrid setup where the AI proposes the extracted record and a human still signs off on higher-value or non-standard trades.

4.2 Reconciliation

Reconciliation is the process of making sure that the firm's own record of its trades, positions, and cash agrees with the records held by every other party in the chain. A wholesale broker needs its internal book to match what the custodian holds, what the prime broker reports, what the exchange or CCP confirms, and what the counterparty thinks happened. When all of these agree, the firm has a reliable picture of where it stands. When they do not, the difference is called a break and somebody has to figure out why.

There are several types of reconciliation that run in parallel. Cash reconciliation checks that the cash balances held at banks match the firm's internal cash ledger. Position reconciliation checks that the firm's record of how many shares of a given security it holds matches what the custodian reports. Trade reconciliation checks that executed trades have been booked correctly and reported to the right downstream systems. Each of these runs daily, sometimes multiple times a day, and each can produce breaks that the operations team has to investigate. Broadridge estimates that reconciliation activities account for 30 to 40% of total back-office labor costs at financial firms [X], although this figure is from their own white paper and should be read as an industry estimate rather than peer-reviewed data.

The reason reconciliation is so labor-intensive is the variety of data sources. The firm's internal booking system speaks one format. The custodian sends statements in another. The prime broker uses yet a different one. SWIFT messages, CSV exports, PDF statements, and proprietary file formats all arrive on different schedules. Before any matching can happen the data has to be normalised, which means parsing it, mapping the fields to a common schema, and handling missing or inconsistent values. Once the data is normalised, traditional reconciliation engines apply rule-based matching: this trade ID

equals that trade ID, this amount within a tolerance of that amount, this date matches that date. Everything that does not match becomes a break, and from there a human investigates.

Reconciliation is one of the areas where AI vendors are most active and where the marketing language is most generous. A closer look at what the tools actually do is worth the effort, because a meaningful share of "AI reconciliation" is a better rule engine with a machine learning layer on top.

The honest assessment is that most reconciliation work in a wholesale broker can be handled with proper IDs and proper rules. If trades carry consistent reference numbers across systems, or if the systems agree on at least a stable suffix, a rule can match them. If the activity in an account is single-purpose, a daily sum compared against the bank statement reconciles it. If the bank fees follow a known fee schedule, the expected total can be calculated and compared directly. None of this needs machine learning. It needs operational discipline and well-designed reference data.

AI starts to genuinely add value in a smaller set of cases. The first is tolerance bands that are not fixed. The research from **schreyer2022anomaly** [X] shows that learned anomaly thresholds outperform fixed rules on general ledger data, and the same principle applies to FX-driven amount differences in reconciliation. A learned tolerance per currency pair or per counterparty catches more real matches than a single band applied across the board. The second is one-to-many groupings where there is no shared reference, for example one invoice covering many trades with no per-trade breakdown. A model trained on historical groupings can propose the grouping where a rule cannot. The third is onboarding new data sources, where the firm does not yet have rules for a particular custodian or product and needs to start matching immediately [X]. The fourth is anomaly detection on already-matched records, where the model flags matches that the rule accepted but that do not fit the firm's historical pattern.

Outside those cases the case for AI is weaker than vendors suggest. The research from **kumar2023payment** [X] reports gains from ML in payment reconciliation, but those gains are largest in environments where reference data is poor and rules have not been maintained. In a wholesale operation where IDs are clean and rules are kept current, the marginal benefit of adding machine learning to reconciliation is smaller than the headline numbers imply.

The maturity of this area follows the same uneven pattern seen in the matching subsection. Rule-based reconciliation engines such as Broadridge, Duco, NeoXam, and SmartStream have been in production for decades and are widely used [X]. The AI layer sitting on top of them is newer and the genuine value of that layer depends heavily on the state of the

firm's underlying data. The academic literature on AI applied to securities reconciliation specifically is also thin. Most peer-reviewed work sits in adjacent domains: general ledger anomaly detection, payment reconciliation, and transaction fraud detection. The techniques transfer, but the absence of brokerage-specific research is itself notable and is a point this survey returns to in chapter 5.

4.3 Risk and collateral management

This subsection covers the operational side of risk and collateral handling in a wholesale broker: tracking exposures, processing margin calls, and managing intraday cash and securities flows. The related topic of counterparty credit risk monitoring is part of the client monitoring discussion in chapter 2 and is not repeated here. The split is deliberate. Credit risk monitoring sits next to KYC and client surveillance because it is about understanding the counterparty. Operational risk and collateral handling sit in the back office because they are about moving cash and securities on time.

The operational problem can be described in plain terms. Some trades close almost immediately. If a client buys shares today, the trade settles a day or two later and the relationship is over. Other trades stay open for months or years. A swap, a repo, or a securities loan keeps running, and during that time the value of the trade keeps changing as the market moves.

This is where collateral comes in. While the trade is open, the side with the loss has to post collateral, which is cash or high-quality securities held by the other side as protection in case of a default. The amount of collateral is recalculated every day based on the current value of the trade. If the market has moved overnight, more collateral has to be posted. This is called a margin call, and there is usually only a few hours, sometimes until end of day, to deliver the assets [X].

A wholesale broker operating in an agency model does not hold these positions itself. The client holds the position. The broker handles the operational side of the process on the client's behalf: calculating what is owed, instructing the movement of assets, reconciling the positions afterwards, and making sure cash and collateral are in the right account at the right time. The broker is not exposed to the market move, but it is exposed to operational risk. A missed margin call, an incorrect calculation, or a delayed settlement instruction can still cost the firm money in penalties, reputational damage, or client claims.

There are two operational areas where AI is starting to appear in this work.

The first is intraday liquidity and cash flow forecasting. A wholesale broker has cash moving in and out of its accounts throughout the day: client settlements, broker fees, custody

charges, FX conversions, and so on. The treasury team needs to know whether cash will be available when an outgoing payment is due. The research from **singh2024liquidity** [X] applies machine learning to liquidity forecasting and shows that ML-based forecasts outperform traditional statistical methods, particularly when transaction patterns are non-linear. The same approach is used by banks for intraday cash flow prediction, where the goal is to estimate the cash position at each hour of the day so that payments can be released or held back accordingly [X]. This is a time-series prediction problem with a clear ground truth, which is the realised cash flow, and it is the kind of problem where ML tends to do well.

The second is margin call processing. A typical wholesale broker receives margin call notifications from multiple counterparties and custodians, each in their own format. Bloomberg messages, SWIFT messages, PDF statements, and emails all arrive in the same morning window. The broker has to read each one, identify which client account it applies to, verify the figure against its own calculation, and instruct the settlement. NLP-based extraction tools, the same family of tools described in the trade confirmation subsection, are being applied here too. The broker still validates and signs off, but the first pass of reading and routing the notice is handled by software [X].

Both of these are useful but neither is dramatic. Liquidity forecasting is an improvement over manual or spreadsheet-based forecasting. Margin call processing is an improvement over manual reading of incoming notices. In both cases the AI sits on top of an existing process rather than replacing it.

The academic literature in this area is also limited for the agency case. Most of the published work on machine learning in margin and collateral is on the principal side and assumes the firm runs its own derivatives book. The papers on initial margin approximation **maInitialMargin2019 mostafa2021im** [X] are about banks calculating SIMM for their own portfolios, not about brokers processing margin calls on behalf of clients. The result is that the AI value proposition for an agency wholesale broker, while real, is narrower than the headline literature suggests. It mostly comes down to forecasting cash and reading documents faster.

5. Risks and Limitations

Up to now this survey has described where AI is being applied in wholesale brokerage. In this chapter we will discuss where the technology falls short, and where its deployment creates risks that did not exist before.

A theme runs through several of the earlier chapters: AI tooling is real but often less mature than vendor marketing suggests. The reconciliation subsection in chapter 3 made

this point explicitly. Four risks are discussed, each specific to the wholesale context. The order moves from the industry wide picture down to the level of an individual procurement decision.

5.1 Vendor concentration and lock-in

Most wholesale firms access AI through third-party vendors rather than building it themselves [X]. Large banks with significant engineering capability build internally for some use cases, but for the majority of wholesale firms, particularly mid-sized agency brokers, the AI capability described in chapters 2 and 3 reaches them through vendor products. This is true across the range of use cases: KYC tooling, document parsing, reconciliation, margin processing, and intraday liquidity forecasting.

The list of vendors is short. For KYC and onboarding, names like ComplyAdvantage, NICE Actimize, Refinitiv, and LexisNexis dominate [X]. For reconciliation, Broadridge, Duco, NeoXam, and SmartStream cover most of the market [X]. For trade confirmation, Xceptor and DTCC's ITP services hold most of the wholesale flow [X]. The underlying LLMs that power the newer AI features come from an even smaller group: OpenAI, Anthropic, Google, with some open-source models from Meta in the mix. A wholesale firm running modern AI tooling is therefore three layers deep in dependencies, and each layer is concentrated in its own right.

This creates several risks. The first is operational. If a vendor has an outage or a data breach, the firm cannot easily switch to a competitor because the integration cost is high and the data formats are not interchangeable. The second is commercial. A vendor with limited competition can raise prices, change licensing terms, or deprecate features the firm relies on. The third is systemic. If a single vendor has an outage across its client base, the wholesale market loses a function at the same time across many firms. This is a different shape of risk from the vendor concentration that already exists in market data (Bloomberg, Refinitiv) because the AI vendor is making decisions, not just providing information.

Regulators have started to notice this. The EU's Digital Operational Resilience Act (DORA), which applies from 17 January 2025, establishes an oversight framework for critical third-party ICT providers in financial services [13]. Whether DORA and similar frameworks are enough to address AI-specific concentration is something this survey returns to in chapter 5.

5.2 Hallucination and confident wrong answers

The most distinctive failure mode of modern AI, particularly LLM-based systems, is the production of outputs that are confidently stated but factually wrong. The model has no

reliable internal signal that it is uncertain, so a wrong answer arrives in the same confident tone as a correct one [12].

In a wholesale context the cost of being wrong is large because trade sizes are large. Chapter 2 raised this point about client-facing chatbots. The same problem appears across most of the AI use cases in chapter 3. An NLP model extracting trade details from a confirmation email might misread a settlement date by a day. An LLM generating a break narrative might attribute the difference to the wrong cause. A document parsing tool might pull a notional figure from the wrong section of a PDF. In each case the output looks correct on the surface, and a human reviewer working through hundreds of items in a morning may not catch the error.

State-of-the-art LLMs have non-trivial hallucination rates on structured information extraction tasks, and that the rate increases when the input format differs from what the model has been trained on. Wholesale documents are not standardised. Each counterparty has its own template and each custodian its own statement layout. This is exactly the condition under which hallucination is most likely.

The mitigation in practice is to keep a human in the loop. Most firms run AI tools as assistants rather than autonomous agents. The AI proposes a value, a human signs off, the work gets done. This is a reasonable approach but it limits the efficiency gain. The AI cannot fully replace the operations analyst because the analyst is still required to verify each output. The headline "AI processes confirmations automatically" needs the footnote "with human review of every record above a certain threshold". The same pattern appeared in the reconciliation and margin call discussions in chapter 3.

5.3 Opacity and accountability gaps

A question in any regulated process is who is responsible when something goes wrong. The traditional answer involves a chain of human accountable owners: an analyst makes a decision, a manager approves it, a compliance officer reviews it, and the record of who decided what is preserved for the regulator. AI complicates this chain.

The opacity issue is twofold. The first is technical. Modern ML models, especially deep learning models, are not easily interpretable. A model trained to detect suspicious transactions can flag a counterparty as high-risk, but explaining exactly why, in terms a compliance officer can defend to a regulator, is harder than for a rule-based system, since deep-learning models are widely characterised as opaque "black boxes" [14]. The second is procedural. Even when an explanation can be produced, the question of who is accountable for the decision becomes blurred. The data scientist who trained the model, the vendor who built it, the operations analyst who acted on its output, and the firm's compliance function are all involved. None of them has full visibility into the others.

This matters because regulators expect a named human owner for material decisions. Under the UK Financial Conduct Authority's Senior Managers Regime and similar frameworks in the EU, individuals are personally accountable for the decisions taken under their function [X]. If an AI declines a counterparty incorrectly and the firm is sued, the accountable manager has to explain the decision. "The model said so" is not a defence. The same applies in reverse. If an AI fails to flag a counterparty that should have been flagged, the manager is still accountable for the gap.

What this means is that the governance frameworks around AI in wholesale brokerage are not yet mature. Most firms have a model risk management function inherited from the credit and market risk side of the business, but applying that framework to AI tools deployed in operations and compliance is a work in progress. The risk is not that AI cannot be governed. The risk is that the gap between deployment speed and governance speed is widening. AI tools are being put into production faster than the policies, sign-off processes, and audit trails around them are being built.

5.4 The gap between vendor marketing and actual capability

The final risk is the gap between the marketing and the actual capability. A meaningful share of what is sold as "AI" in wholesale brokerage is not AI in the modern sense. It is configurable rule engines, fuzzy string matching, optical character recognition, and statistical models that have existed for decades, packaged with a UI and marketing language that suggests machine learning[X].

This was discussed in the reconciliation subsection in chapter 3 and the pattern generalises. A vendor product labeled "AI-powered" may include a small ML layer on top of an otherwise traditional system[X]. The label is commercially valuable. Firms want to be seen to be adopting AI. Vendors want to sell at AI prices. The result is that vendors over promise and firms adopting AI underdeliver[X].

The risk in this is not that the tools are not working[X]. The risk is that firms make decisions based on these systems. For example an entire process could be redesigned because firms expect that machine learning will improve the process. But only discover the gap between marketing and reality after it fails or improvements can not be seen in numbers.

6. Regulatory Landscape

7. Gap analysis

8. Conclusion